

NBA Player Salaries: Who's Overpaid and Underpaid in 2025

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This is a brief description of NBA player salaries and determining players who are over or underpaid in 2025 based on on-court performance using machine learning models. Using data from Kaggle.

Introduction

The NBA is one of the most successful businesses in sports, and within it are some of the highest paid athletes in the world. NBA teams pay players millions of dollars, but some may be over or undervalued relative to their performance. This brief examines which players are overpaid or underpaid in 2025 while identifying the key factors that determine salaries.

Data & Methods

The analysis used NBA player statistics and salary data from 2000 – 2025, from a Kaggle dataset ([NBA Player Stats and Salaries 2000-2025](#)).

A Random Forest Regression model was trained on all historical data prior to 2025 to predict player salaries based on on-court performance metrics, including:

- Points scored (PTS)
- Age
- Year: *Captures salary cap inflation/league revenue growth*
- Minutes played (MP)
- Field goals made (FG)
- Defensive rebounds (DRB)
- Assists (AST)
- Turnovers (TOV)
- Games started (GS)
- Two-point field goals made (2P)

Salary discrepancies were calculated as:

Difference = Actual Salary - Predicted Salary

Percent over/underpaid was calculated as:

Percent Over/Underpaid = (Difference / Predicted Salary) * 100

For visualization and interpretation purposes, the 2025 dataset was filtered to exclude players earning less than \$1.1M to focus only on players on standard NBA contracts.

Players are then sorted to identify the top 10 overpaid and underpaid, as shown in Figure 1, and the top 5 players are shown in Table 1.

The Random Forest model performed well, reporting an R^2 of 0.75 and a root mean squared error of \$6,233,320 indicating a good predictive accuracy.

Results & Visualization

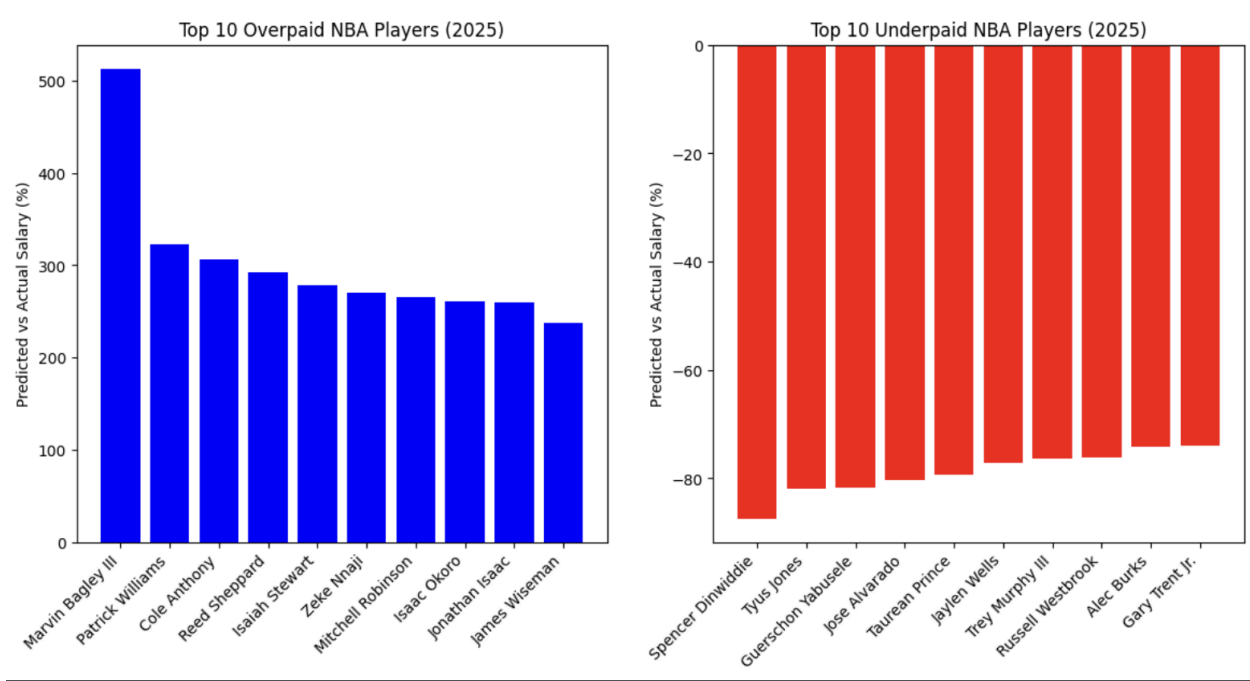


Figure 1. Top 10 Overpaid and Top 10 Underpaid NBA players in 2025.

Figure 1 shows the largest discrepancies between actual and predicted salaries, highlighting players who earn far above or below their predicted values. The side by side bar chart allows a clear comparison of the top 10 overpaid and underpaid NBA players in 2025.

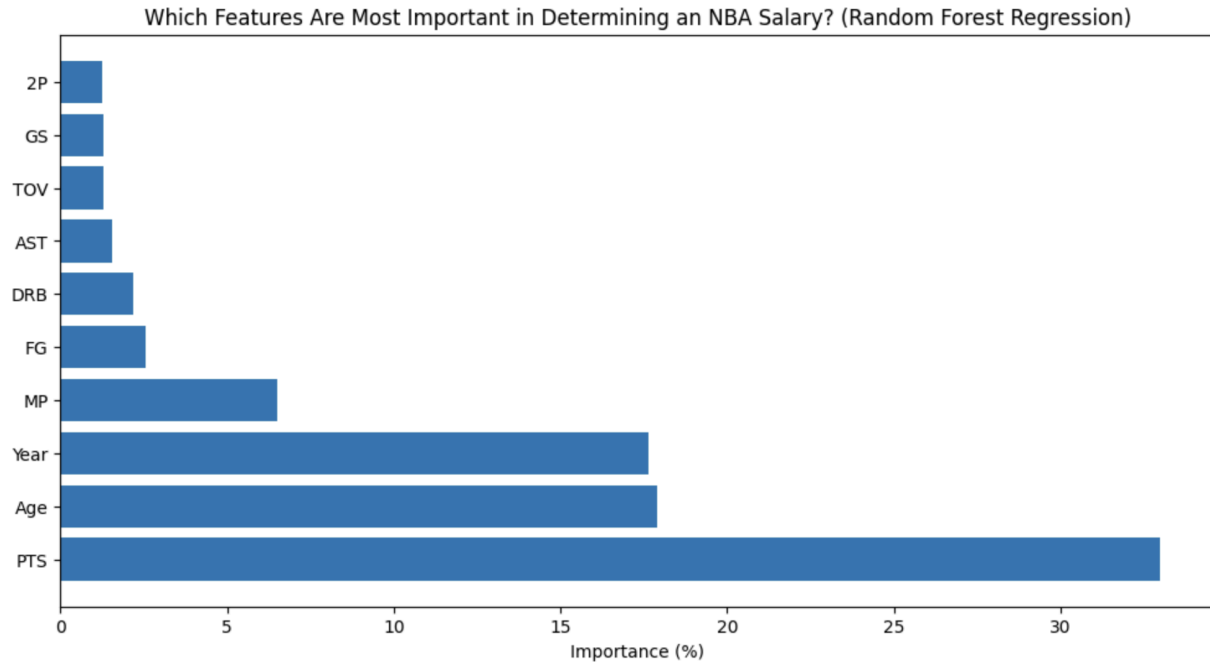


Figure 2. Key Factors Influencing NBA Player Salaries in 2025.

Figure 2 illustrates the key factors influencing NBA player salaries according to the machine learning model. Points scored (PTS), player age, and year the salary was paid are the most influential predictors followed by minutes played (MP) and field goals (FG).

Player	Actual Salary (\$)	Predicted Salary (\$)	Difference (\$)	Over/Underpaid (%)
Marvin Bagley III	12,500,000	2,039,059	10,460,941	513.0
Patrick Williams	18,000,000	4,262,950	13,737,050	322.2
Cole Anthony	12,900,000	3,176,385	9,723,615	306.1
Reed Sheppard	10,098,960	2,570,906	7,528,054	292.8
Isaiah Stewart	15,000,000	3,961,091	11,038,909	278.7
Spencer Dinwiddie	2,087,519	16,581,021	-14,493,502	-87.4
Tyus Jones	2,087,519	11,509,720	-9,422,201	-81.9
Guerschon Yabusele	2,087,519	11,448,980	-9,361,461	-81.8
Jose Alvarado	1,988,598	10,131,857	-8,143,259	-80.4
Taurean Prince	2,087,519	10,094,532	-8,007,013	-79.3

Table 1. Top 5 Overpaid and Underpaid NBA Players by salary discrepancy in 2025.

Table 1 provides the precise dollar and percentage differences for the top 5 overpaid and underpaid players in 2025. Negative values (red) indicate underpayment relative to predicted salary, while positive values indicate overpayment.

Discussion

Limitations include the exclusion of off-court factors such as off-court behavior, locker room presence, and player potential. Additionally, the analysis does not account for injuries, contract clauses, or endorsement deals, all of which can influence salaries. Team strategy, market size, and player reputation may also affect pay separate from off court performances. The Random Forest model predicts well but cannot fully account for these aspects. Some moderately performing players who are overpaid suggest teams value leadership, mentorship, or long term growth beyond raw stats.

When examining overpayment by position we notice power forwards and small forwards are the most frequently overpaid at 61.8% and 61.0%. Point guards and centers are slightly overpaid at 55.4% and 53.1%. Shooting guards are most often underpaid with 54.7% earning less than predicted. This finding indicated that forward positions tend to be valued more highly relative to performance while some guards provide greater value than their salary suggests.

Conclusions

Machine learning provides insights into NBA salary dynamics, highlighting over and undervalued players based on performance statistics. Points scored, age, and experience are the strongest predictors, while other stats play a smaller role. This analysis can help teams, analysts, and agents evaluate a player's value, identify market inefficiencies, and make informed contract decisions.

Teams should leverage this discrepancy model not just to flag overpaid players, but to target high-performing players on insufficient contracts to pursue trades for such as adding undervalued shooting guards to their roster. Agents can utilize the model's predicted salary as a benchmark for future contract negotiations.